

# A Unified Edge-AI Framework for Real-Time Multi-Person Face Recognition and Adaptive Identity Mining in Smart Agriculture

**Abstract**—Smart agriculture environments require intelligent monitoring systems to ensure secure farm access, worker accountability and operational transparency. However, existing surveillance solutions rely heavily on cloud-based processing, leading to latency, privacy risks and dependence on continuous internet connectivity, which limits their applicability in rural and low-connectivity regions such as India. To address these challenges, this paper proposes a unified Edge-AI-based computer vision framework for real-time multi-person face recognition and identity-aware monitoring. The system integrates YOLOv8-based face detection, FaceNet-based embedding extraction, cosine similarity-based recognition and DeepSORT tracking within a single on-device pipeline, along with an adaptive identity mining mechanism to track recurring unknown individuals. The framework is deployed on an NVIDIA Jetson Orin Nano edge device, selected for its balance of computational efficiency, low power consumption and suitability for real-time edge inference. Experimental results demonstrate that the system achieves 97.5% accuracy, 97.8% mAP, 96.4% IDF1 and 94.6% MOTA, while maintaining real-time performance at 25 FPS. The proposed approach provides a cost-effective, privacy-preserving and scalable solution for intelligent farm monitoring in resource-constrained environments.

**Index Terms**—Edge-AI, Computer Vision, Face Recognition, Person Re-Identification, Multi-Object Tracking, Adaptive Identity Mining

## I. INTRODUCTION

Smart farming is very important for keeping rural economies going in developing countries where raising animals for food is a major source of income. But conventional surveillance systems have serious problems when it comes to keeping an eye on workers, making sure only authorized people can get into the farm and finding people who shouldn't be there. Most current solutions use cloud-based processing, which adds latency, depends on bandwidth and raises privacy concerns. This makes them not work well in rural areas with poor internet connections [1], [2]. Edge intelligence allows processing to happen on the device in real time, which cuts down on communication overhead while improving data privacy and operational efficiency [3]. Smart agriculture monitoring systems have also shown a lot of promise for improving farm management and surveillance [4], [5], which has led to a greater need for efficient and easy-to-use edge-based solutions.

Recent improvements in computer vision have made it possible to make systems that can efficiently detect, recognize and track objects that can be used at the edge. YOLO and other lightweight object detection models can accurately and quickly get visual information from video streams [6], [7]. Recognition

and person re-identification techniques based on embedding create strong feature representations for matching identities in different environments [8]–[10]. Multi-object tracking methods make sure that identities stay the same over time by linking them across frames [11], [12]. Edge computing architectures make it easy to install these kinds of systems on devices with limited resources [13].

Even with these improvements, most current methods still treat detection, tracking and recognition as separate parts, which makes them less useful for continuous identity-aware monitoring. Also, they don't do a good job of solving the problem of finding and keeping track of unknown people who keep coming back.

To overcome these constraints, this paper introduces a cohesive Edge-AI-based framework for real-time multi-person face recognition and surveillance in smart agricultural settings. The suggested system combines face detection, embedding-based recognition and multi-object tracking into a single on-device pipeline that runs on an NVIDIA Jetson Orin Nano. An adaptive identity mining mechanism is also introduced to find and keep track of people who keep coming back. The system also supports one-shot identity recognition, which means that you can reliably identify someone again using just one reference image, even when the time and place are different.

Our Proposed work's main contributions are the creation of a unified Edge-AI pipeline that combines face detection, multi-object tracking and identity recognition for real-time monitoring, as well as an adaptive identity mining mechanism for dealing with people who keep coming back. The system is also deployed and tested in real time on an NVIDIA Jetson Orin Nano edge device and evaluated on a smart farm dataset in the real world under tough conditions.

## II. RELATED WORK

Recent improvements in computer vision have made it much easier to detect and track multiple individuals. YOLO-based detection frameworks are widely used due to their real-time performance and efficiency, making them suitable for dynamic environments [14], [15]. Multi-object tracking approaches, including deep association-based tracking and embedded AI tracking systems, enable identity-aware tracking by linking detections across frames [16]. Recent advancements in tracking and data association techniques further improve tracking reliability. However, these approaches either increase

computational complexity or focus primarily on tracking performance without integrating identity recognition. As a result, these methods are less suitable for unified real-time monitoring on resource-constrained edge devices.

Face recognition and person re-identification techniques have significantly improved identity-aware systems by enabling robust matching of individuals across varying conditions. Embedding methods based on deep learning create feature representations that improve recognition accuracy when there are problems like occlusion, changes in pose and changes in lighting [17], [18]. These methods are often tested on their own, without detection and tracking modules, even though they work well. As a result, many systems don't have smooth integration between tracking and recognition, which makes pipelines that aren't good for continuous real-world monitoring.

Edge computing has made it possible to use real-time vision systems with less lag time and better performance. Recent research indicates that lightweight deep learning models can function directly on embedded platforms without dependence on cloud infrastructure [19], [20]. These methods make the system more responsive and cut down on communication overhead. Most systems that are already in place, on the other hand, don't fully support identity-aware monitoring and mostly focus on tracking or detecting tasks that are separate from each other. Cloud-based vision systems also pose privacy risks because they send sensitive visual data over the internet. This shows how important it is to have fully on-device processing frameworks.

Vision-based monitoring systems have also been explored in agricultural applications, demonstrating the feasibility of AI-driven solutions in real farm environments [21], [22]. However, these systems mainly focus on environmental monitoring or activity detection rather than identity-aware surveillance. Furthermore, they do not address the challenge of recognizing recurring unknown individuals or maintaining long-term identity records.

Overall, no existing edge-based framework integrates detection, tracking, recognition and unknown identity management into a single pipeline. The proposed system is motivated by these limitations and addresses the need for real-time, identity-aware and privacy-preserving monitoring in smart agriculture environments.

### III. MATERIALS AND METHODS

#### A. Materials

1) *Dataset Collection*: The dataset used for training and testing was collected using a Zebronics FHD USB webcam capturing video at 25 FPS. Data was recorded at a resolution of  $1920 \times 1080$  pixels from a real cattle farm. The dataset consists of approximately 14,000 face images, including ten known identities (authorized personnel and farm workers) and twelve unknown identities observed during deployment.

Changes in illumination, motion blur, occlusion and posture were among the challenging real-world conditions under which the data was collected. The facial regions extracted

from frames were manually annotated. The dataset was split into training, validation and testing sets in the ratio of 70%, 20% and 10%, respectively. While the frequency of unknown identities varies in real-world scenarios, there are approximately 600–700 samples for every known identity. The dataset includes both male and female participants between the ages of 20–60.

Even though the dataset isn't very big, it was made to test how well the system works on edge devices with limited resources in real-world situations. The results show that the system works well, even though there are some problems. All data collection was done with the participants' knowledge and consent, which made sure that their privacy was protected and that ethical standards were followed.



Fig. 1. Sample face images under varying conditions such as illumination, pose, blur and occlusion.

#### B. Methods

1) *Face Detection*: The YOLOv8n (nano) model was used for face detection because of its lightweight design and edge deployment compatibility. To balance detection speed and accuracy, the model was set up with a Non-Maximum Suppression (NMS) threshold of 0.4 and a confidence threshold of 0.5.

2) *Multi-Object Tracking*: The DeepSORT algorithm, which uses deep appearance features and Kalman filtering for identity association, was used to track detected faces. Because DeepSORT is more computationally efficient than more sophisticated trackers, it was chosen for real-time edge deployment.

3) *Face Embedding Extraction*: For feature extraction, a FaceNet-based embedding model with a 512-dimensional output was employed. To enhance domain-specific performance

under farm conditions, the model was refined on the gathered dataset after being initialized with a pretrained InceptionResNetV1 architecture.

4) *Identity Recognition*: Identity recognition is performed using cosine similarity between extracted embeddings and stored identity embeddings. A similarity threshold  $\tau = 0.6$  is empirically selected to balance false positives and false negatives. If the similarity score exceeds  $\tau$ , the identity is classified as known otherwise, it is labelled as unknown. The embedding-based representation allows for one-shot identification, which means that the system can recognize people using just one reference image and then reliably identify them again at different times and in different lighting and pose situations.

5) *Adaptive Identity Mining*: An adaptive identity mining mechanism was put in place to deal with people whose identities are not known. Using similarity-based grouping, embeddings of unknown faces are stored and grouped together over time. When a cluster hits a certain frequency threshold, it gets a temporary identity label. This makes it possible to keep track of people who visit more than once and keep records of their visits over time.

The clustering of unknown identities is performed using a similarity-based threshold grouping mechanism, where embeddings with cosine similarity above a predefined threshold are grouped into the same identity cluster. To ensure efficient operation over time, a memory management strategy is employed in which older or less frequently observed identity embeddings are periodically pruned, preventing memory overflow and maintaining system scalability.

#### IV. IMPLEMENTATION

##### A. Experimental Setup

The suggested edge-based monitoring system was deployed on a real cattle farm and evaluated under varying conditions, including changes in lighting and subject movement. It demonstrated reliable detection, recognition, and tracking performance during operation. The system was implemented on an NVIDIA Jetson Orin Nano with an 8-core CPU, 8 GB RAM, and an integrated GPU for efficient deep learning inference. A USB camera with a resolution of  $1920 \times 1080$  at 25 FPS was installed at the farm entrance to capture the movement of workers and visitors. Video streams are processed entirely on-device using an edge-first architecture, eliminating the need for cloud infrastructure and ensuring low latency, reduced bandwidth usage, and enhanced data privacy. The processing pipeline includes face detection, embedding extraction, identity recognition, and multi-object tracking, along with an event-driven alert system via a Telegram interface. The average end-to-end latency from detection to alert notification is approximately 1.2 seconds under normal operating conditions.

The Fig. 2, illustrates the hardware setup for edge-based deployment, including the NVIDIA Jetson Orin Nano, USB camera, display, and input devices. The system is configured for real-time video processing and continuous operation.



Fig. 2. Hardware setup for edge-based deployment including Jetson Orin Nano and camera system.

##### B. Methodology

As shown in Fig. 3, the system pipeline begins with video acquisition, followed by preprocessing and face detection using YOLOv8. Extracted face embeddings are passed to the identity recognition module, while DeepSORT ensures temporal consistency. Unknown identities are processed through the adaptive identity mining module, enabling clustering and long-term tracking.

1) *Video Stream Processing*: Let the captured video stream be defined as:

$$V = \{F_t \mid t = 1, 2, \dots, T\} \quad (1)$$

where  $F_t$  represents the frame at time  $t$ .

The video stream consists of a sequence of frames:

$$V = \{F_t\}_{t=1}^T \quad (2)$$

To reduce computational load while preserving temporal information, Frames are sampled at a fixed interval ( $k = 2$ ) to reduce computational load.

$$F_t^s = F_t, \quad \text{where } t \bmod k = 0 \quad (3)$$

where  $F_t^s$  represents the sampled frame.

2) *Face Detection*: Face detection is performed using a YOLOv8-based model:

$$D(F_t) = \{(b_i, c_i)\}_{i=1}^N \quad (4)$$

where  $b_i$  denotes the bounding box coordinates and  $c_i$  represents the detection confidence score for the  $i$ -th detected face.

3) *Face Embedding Extraction*: For each detected face, a 512-dimensional embedding is extracted using a FaceNet-based model:

$$f_i \in \mathbb{R}^{512} \quad (5)$$

where  $f_i$  represents the feature embedding of the  $i$ -th face.

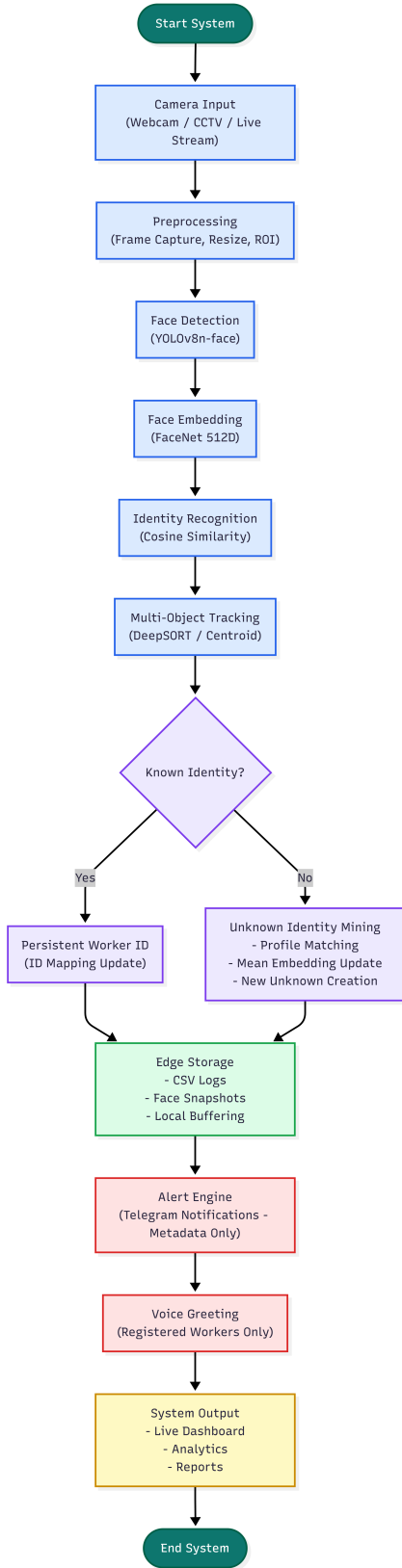


Fig. 3. System architecture showing on-device face detection, embedding-based recognition, multi-object tracking and adaptive identity mining pipeline.

4) *Identity Recognition*: Identity matching is performed using cosine similarity:

$$S(f_i, f_j) = \frac{f_i \cdot f_j}{\|f_i\| \|f_j\|} \quad (6)$$

where  $f_i$  and  $f_j$  denote embedding vectors of two faces and  $S(\cdot)$  denotes the cosine similarity function.

If  $S(f_i, f_j) > \tau$ , where  $\tau = 0.6$  is a predefined similarity threshold, the identity is classified as known. otherwise, it is labeled as unknown. The embedding-based design enables one-shot identity recognition, allowing reliable re-identification using a single reference image under varying temporal and environmental conditions.

5) *Multi-Object Tracking*: Temporal identity association is performed using DeepSORT:

$$x_t = Ax_{t-1} + w_t \quad (7)$$

where  $x_t$  represents the object state at time  $t$ ,  $A$  denotes the state transition matrix and  $w_t$  represents process noise.

6) *Alert Generation*: An alert is triggered when an individual is detected:

$$\text{Alert} = \begin{cases} 1, & S(f_i, f_j) > \tau \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

where the alert flag indicates the presence of a detected individual.

### C. Evaluation Metrics

System performance is evaluated using standard metrics for detection, recognition and tracking.

1) *Re-Identification Accuracy*:

$$A_{reid} = \frac{N_{correct}}{N_{total}} \quad (9)$$

where  $N_{correct}$  denotes the number of correctly identified instances and  $N_{total}$  denotes the total number of evaluated instances.

2) *Mean Average Precision (mAP)*:

$$mAP = \frac{1}{C} \sum_{c=1}^C \int_0^1 P_c(r) dr \quad (10)$$

where  $C$  denotes the number of classes and  $P_c(r)$  represents the precision-recall curve for class  $c$ .

3) *IDF1 Score*:

$$IDF1 = \frac{2 \times IDTP}{2 \times IDTP + IDFP + IDFN} \quad (11)$$

where  $IDTP$ ,  $IDFP$  and  $IDFN$  denote identity true positives, false positives and false negatives, respectively.

4) *Multi-Object Tracking Accuracy (MOTA)*:

$$MOTA = 1 - \frac{FN + FP + IDS}{GT} \quad (12)$$

where  $FN$ ,  $FP$  and  $IDS$  denote false negatives, false positives and identity switches and  $GT$  denotes the total number of ground-truth objects.

### 5) FPS (FPS):

$$FPS = \frac{N_{frames}}{t_{end} - t_{start}} \quad (13)$$

where  $N_{frames}$  is the number of processed frames and  $(t_{end} - t_{start})$  is the total processing time.

## V. RESULTS AND DISCUSSION

This section presents a comprehensive evaluation of the proposed Edge-AI computer vision framework for real-time multi-person recognition and monitoring in smart farm environments. The analysis includes training performance, real-world deployment results, ablation studies and comparative evaluation.

### A. Training Results

The face recognition training performance shows that it can learn features well and converge steadily. The training loss drops quickly at first and then levels off, while the top-1 accuracy keeps going up and gets close to the best performance. The validation loss goes up and down in a similar way, which means that the model generalizes well without overfitting.

The convergence process is illustrated in Fig. 4. It demonstrates that the learned embeddings are robust enough to deal with variations in motion blur, occlusion and lighting. Strong identity classification performance with low inter-class confusion is shown by the confusion matrix in Fig. 5. This suggests that even in difficult visual situations, the learned embeddings are very discriminative.

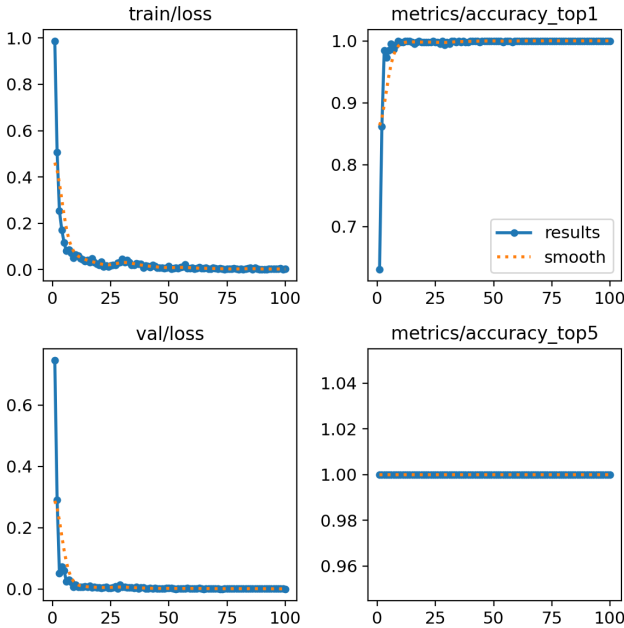


Fig. 4. Training and validation performance curves.

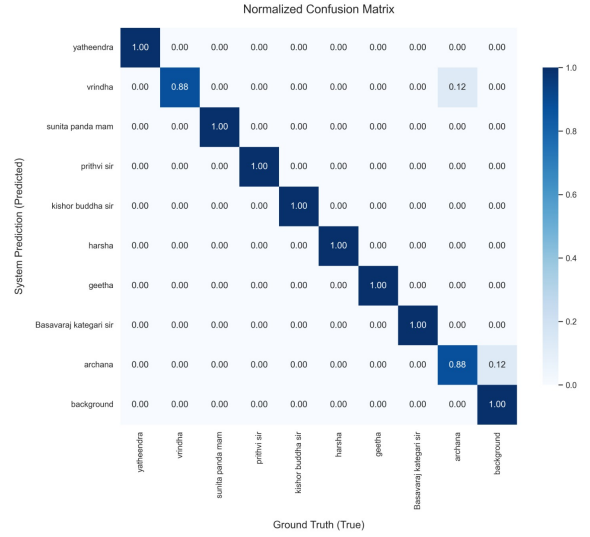


Fig. 5. Normalized confusion matrix illustrating identity recognition performance across multiple registered individuals.

### B. Field Trial Results

The proposed system was deployed on an NVIDIA Jetson Orin Nano device in a cattle farm environment to evaluate its performance in practical situations. As it continuously processes live video streams, the system performs the entire vision pipeline on-device, including face detection, embedding extraction, identity recognition and multi-object tracking. The deployment scenario includes challenging real-world scenarios such as multiple people appearing at once, motion blur, partial occlusions and variations in illumination.

Fig. 6 displays the results of real-time multi-person recognition, where individuals are accurately identified and labeled with identity information. The integration of DeepSORT ensures temporal consistency by reducing identity switches and maintaining steady tracking between frames. Fig. 7 illustrates how the pipeline is executed on the Jetson platform, showing efficient real-time processing. In Fig. 8, which shows the actual installation of the system in the farm setting, the camera is set up to observe the entry region.

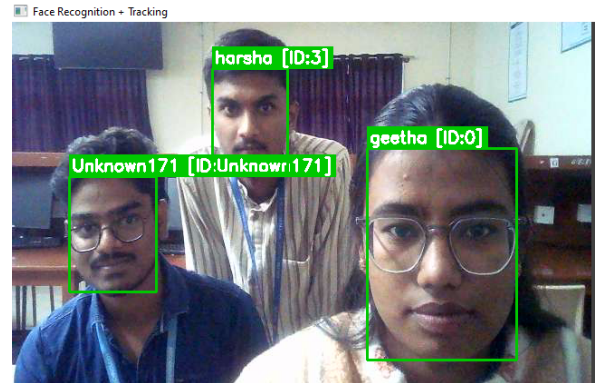


Fig. 6. Real-time multi-person recognition and tracking output generated by the edge deployment system.



Fig. 7. Real-time edge deployment of the proposed monitoring system on the NVIDIA Jetson Orin Nano platform performing face detection, identity recognition and multi-person tracking using a USB camera.



Fig. 8. Real-world deployment of the edge-based monitoring system in a cattle farm environment

The event-driven alert mechanism generates real-time notifications via a Telegram interface, as shown in Fig. 9. The average end-to-end latency from detection to alert delivery is approximately 1.2 seconds, enabling timely situational awareness for farm operators. Additionally, the mobile dashboard interface shown in Fig. 10. enables remote monitoring, allowing users to access real-time system outputs, identity logs and alert information.

### C. Performance Analysis

The system achieves an overall re-identification accuracy of 97.5%. Performance under different conditions is summarized as follows:

- Normal conditions: 98.2%
- Illumination variation: 96.8%
- Partial occlusion: 95.9%
- Motion blur: 95.3%

These results indicate that the model maintains high robustness under real-world variations, although performance slightly degrades under severe occlusion and blur.

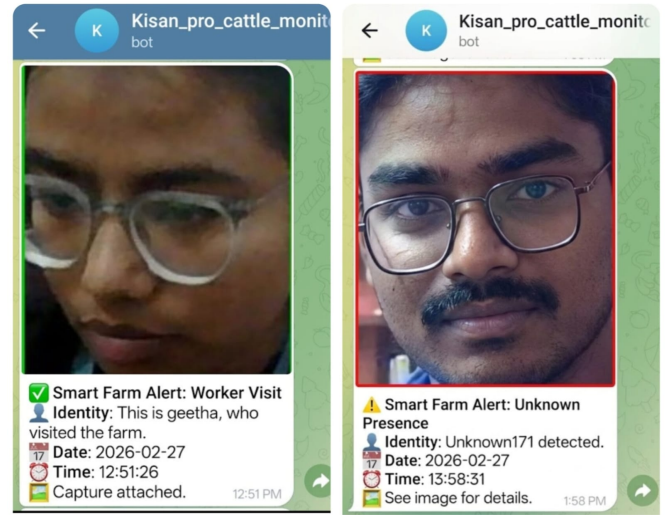


Fig. 9. Real-time Telegram alert notifications generated by the system for registered individuals and unknown visitors, enabling identity-aware monitoring and real-time situational awareness.

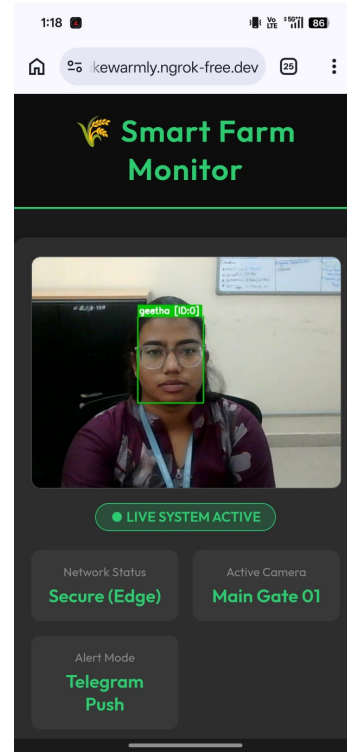


Fig. 10. Real-time mobile dashboard interface displaying live monitoring results from the smart farm system. The dashboard shows detected identity, system status, camera information, and alert mode through a remote static web link, enabling continuous monitoring and remote accessibility.

### D. Comparative Analysis

The qualitative comparison between the proposed model and other traditional models of monitoring that are vision-based has been provided in Table I. It is extremely difficult to provide any comparison in this regard due to the presence

of variations in the dataset, model design and testing protocol among others. As such, the figures provided above should be taken as recommendations only.

TABLE I  
QUALITATIVE PERFORMANCE COMPARISON WITH REPRESENTATIVE  
METHODS FROM LITERATURE

| Method                          | Dataset                | Acc. (%)    | mAP (%)     | IDF1 (%)    | MOTA (%)    |
|---------------------------------|------------------------|-------------|-------------|-------------|-------------|
| YOLO-based Detection [14], [15] | WIDER FACE / Custom    | 94–96       | 92–94       | –           | –           |
| Tracking (DeepSORT) [16]        | MOT17                  | –           | –           | 92–94       | 90–92       |
| Re-ID Methods [17], [18]        | Market-1501 / DukeMTMC | 95–96       | 94–95       | 92–93       | –           |
| Edge-based Systems [19]–[22]    | Custom                 | 95–97       | 94–96       | 93–95       | 91–93       |
| <b>Proposed (Ours)</b>          | Custom Farm            | <b>97.5</b> | <b>97.8</b> | <b>96.4</b> | <b>94.6</b> |

While YOLO-based approaches are efficient in detection, they do not manage tracking identities. Conversely, tracking algorithms make it easier for users to link detected objects without identifying them. While Re-ID approaches enhance feature matching, they are evaluated separately most of the time. Modern edge-based approaches offer high efficiency however, they lack a unified solution approach.

Unlike previous works, the suggested approach includes detection, recognition and tracking in one edge-based system. Furthermore, adaptive identity mining is added to the algorithm to cope with frequently recurring identities. Such an advanced framework enables real-time monitoring of identities in a practical way.

#### E. Discussion

The experimental findings have indicated that integrating detection, recognition and tracking into a single pipeline increases the overall effectiveness of the entire framework. The edge computing technology allows processing information without relying on cloud infrastructure in real-time, thus providing low latency and maintaining user privacy. The framework demonstrates that it is possible to achieve successful re-identification of the target individual utilizing limited reference data within various time intervals and adverse environmental conditions. However, several drawbacks need to be addressed. The system may perform worse in the presence of crowded scenes, bright lighting and occlusions. Moreover, implementing support for multi-camera systems may require additional efforts from the developers. The system maintains a stable rate of 25 FPS through employing a lightweight YOLOv8 algorithm, frame subsampling and GPU acceleration in the Jetson environment. These design decisions guarantee a proper balance between accuracy and computational efficiency. Overall, the proposed framework outlines its limitations for future improvements and practical potential for use in smart farm surveillance applications.

#### F. Limitations

It is clear that the suggested approach performs efficiently, yet it faces certain challenges. In cases when the number of people is high, the lighting is bright and occlusions are strong, the performance decreases significantly. It should also be noted that the identity linking framework should be enhanced to make it possible to employ the existing configuration in multichannel environments.

## VI. CONCLUSION AND FUTURE WORK

Our proposed framework shows that a single Edge-AI-based computer vision pipeline can be used to monitor smart farms in real time and with knowledge of who is there. The system gets high accuracy and low latency performance by using face detection, embedding-based recognition and multi-object tracking all on the device, without needing cloud infrastructure. Experimental results show that lightweight deep learning models running on edge hardware can still work well in the real world, even when there are changes in lighting, motion blur and occlusion. They can also process data in real time at 25 FPS. The combination of tracking and embedding-based recognition makes sure that identity association is always the same and the adaptive identity mining mechanism works well with people who keep coming back. The proposed framework achieves 97.5% re-identification accuracy, 97.8% mAP, 96.4% IDF1 and 94.6% MOTA, demonstrating its effectiveness for scalable, privacy-preserving and real-time monitoring in smart agriculture applications.

Future efforts will focus on improving the framework so that it can handle multi-camera setups and cross-camera re-identification for more thorough monitoring. To make things run better and work on low-power edge devices, we will look into model optimization techniques like quantization and pruning. We will develop lightweight spatiotemporal analysis techniques to detect anomalies and comprehend behavior while minimizing additional computational requirements. Federated learning will also be studied to enable farms to collaborate on model enhancement while ensuring data privacy and facilitating edge deployment.

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